~~Ha~~ Mid - Sem (2024)

① (i) $H = \sum p_i \dot{q}_i - L$

For σ: $p_\sigma = \dfrac{\partial L}{\partial \dot{\sigma}} = \dfrac{m\dot{\sigma}}{\sigma^2+\tau^2}$ — (I)

for τ: $p_\tau = \dfrac{\partial L}{\partial \dot{\tau}} = \dfrac{m\dot{\tau}}{\sigma^2+\tau^2}$ — (II)

$\therefore H = p_\sigma \dot{\sigma} + p_\tau \dot{\tau} - L$

$= \dfrac{m\dot{\sigma}^2}{\sigma^2+\tau^2} + \dfrac{m\dot{\tau}^2}{\sigma^2+\tau^2} - \dfrac{m}{2}\left(\dfrac{\dot{\sigma}^2+\dot{\tau}^2}{\sigma^2+\tau^2}\right)$

$$\boxed{H = \frac{m}{2}\left(\frac{\dot{\sigma}^2+\dot{\tau}^2}{\sigma^2+\tau^2}\right)}$$ (Ans) — (III)

(ii) For σ: $\cancel{\dfrac{\partial H}{\partial \dot{\sigma}} = \dfrac{m\dot{\sigma}}{\sigma^2+\tau^2}, \quad \dfrac{\partial H}{\partial p_\sigma} = p_\sigma}$ — (IV)

$$\boxed{H = \frac{\sigma^2+\tau^2}{2m}\left(p_\sigma^2 + p_\tau^2\right)}$$ (from (I) & (II)).

(Ans) for part (a)

$$\therefore \frac{\partial H}{\partial p_\sigma} = \cancel{\frac{\partial H}{\partial \dot\sigma} \cdot \frac{\partial \dot\sigma}{\partial p_\sigma}} = \cancel{\frac{m\dot\sigma}{\sigma^2+\tau^2}} \cdot \frac{\partial \dot\sigma}{\partial p_\sigma} \quad (\sigma^2+\tau^2)\frac{p_\sigma}{m}$$

$$\boxed{\therefore \cancel{\frac{\partial p_\sigma}{\partial \dot\sigma} = \frac{m}{\sigma^2+\tau^2}} \text{ (from (I))} \quad \Rightarrow \cancel{\frac{\partial \dot\sigma}{\partial p_\sigma} = \frac{\sigma^2+\tau^2}{m}}}$$

$$\therefore \frac{\partial H}{\partial \sigma} = \left(\frac{p_\sigma^2 + p_\tau^2}{m}\right)\sigma = -\dot p_\sigma \quad \text{— (IV)}$$

$$\therefore \frac{\partial H}{\partial p_\sigma} = \cancel{\frac{m\dot\sigma}{\sigma^2+\tau^2}} \cdot \frac{\sigma^2+\tau^2}{m} p_\sigma = \dot\sigma \quad \text{— (V)}$$

~~(IV) &~~ (V) ~~are~~ is are the Hamiltonian's EOM for σ.

(IV) &

For τ: Since the Langrangian is symmetric in σ & τ, (changing $\sigma \leftrightarrow \tau$ does not affect the expressions),

$$\frac{\partial H}{\partial \tau} = \frac{\tau}{m}(p_\sigma^2 + p_\tau^2) = -\dot p_\tau$$

$$\boxed{\cancel{\frac{\partial H}{\partial \dot\tau} = \frac{m\dot\tau}{\sigma^2+\tau^2} = p_\tau}}$$

$$\frac{\partial H}{\partial p_\tau} = \dot\tau = \left(\frac{\sigma^2+\tau^2}{m}\right)p_\tau$$

are ~~is~~ are the Hamiltonian's EOM for τ.

(2) (i) $p_1 = \frac{\partial L}{\partial \dot q_1} = 2\dot q_1 + k_2 \dot q_2$ — (I)

$p_2 = \frac{\partial L}{\partial \dot q_2} = 2\dot q_2 + k_2 \dot q_1$ — (II)

(ii) (I) + (II) $\Rightarrow$ $p_1 + p_2 = 2(\dot q_1 + \dot q_2) + k_2(\dot q_1 + \dot q_2)$

$= (\dot q_1 + \dot q_2)(k_2 + 2)$.

$\Rightarrow \dot q_1 + \dot q_2 = \frac{p_1 + p_2}{k_2 + 2}$ — (III)

(II) − (I) $\Rightarrow$ $p_2 - p_1 = 2(\dot q_2 - \dot q_1) + k_2(\dot q_2 - \dot q_1)$

$= (\dot q_2 - \dot q_1)(2 - k_2)$.

$\Rightarrow \dot q_2 - \dot q_1 = \frac{p_1 - p_2}{k_2 - 2}$ — (IV)

$\therefore$ (III) + (IV) $\Rightarrow$ $\dot{q}_2 = \frac{1}{2}\left[\frac{p_1+p_2}{k_2+2} + \frac{p_1-p_2}{k_2-2}\right]$ (Ans)

$\therefore$ (III) − (IV) $\Rightarrow$ $\dot{q}_1 = \frac{1}{2}\left[\frac{p_1+p_2}{k_2+2} - \frac{p_1-p_2}{k_2-2}\right]$ (Ans)

(III) $H = \sum p_i \dot{q}_i - L$

$= (2\dot{q}_1 + k_2\dot{q}_2)\dot{q}_1 + (2\dot{q}_2 + k_2\dot{q}_1)\dot{q}_2 - \dot{q}_1^2 - \dot{q}_2^2 - k_1 q_1^2 - k_2 \dot{q}_1 \dot{q}_2$

$H = \dot{q}_1^2 + \dot{q}_2^2 + k_2 \dot{q}_2 \dot{q}_1 - k_1 q_1^2$ (Ans) (*)

(3) $H(q_i, p_i, t) = \sum p_i \dot{q}_i - L(q_i, \dot{q}_i, t)$ — (a)

$\Rightarrow \frac{dH}{dt} = \sum \frac{\partial H}{\partial q_i}\dot{q}_i + \frac{\partial H}{\partial p_i}\dot{p}_i + \frac{\partial H}{\partial t}$ — (V)

From Hamiltonian's EOM:

$\frac{\partial H}{\partial p_i} = \dot{q}_i$; $\cancel{\frac{\partial H}{\partial \dot{q}_i} = p_i}$ $\frac{\partial H}{\partial q_i} = -\dot{p}_i$

Substituting these into (V):

$\frac{dH}{dt} = \sum \left(\frac{\partial H}{\partial q_i}\right)\left(\frac{\partial H}{\partial p_i}\right) - \left(\frac{\partial H}{\partial p_i}\right)\left(\frac{d}{dt}\left(\frac{\partial H}{\partial \dot{q}_i}\right)\right) \cdot \left[\frac{\partial H}{\partial t} = 0\right]$

~~Now,~~ $\cancel{\frac{d}{dt}\left(\frac{\partial H}{\partial \dot{q}_i}\right) = \frac{\partial H}{\partial \dot{q}_i}}$ $= 0$. (Proved)

(*) $\dot{q}_2 = \frac{1}{2}\left[\frac{\cancel{k_2 p_1 + k_2 p_2} (k_2 - 2)(p_1+p_2) + (k_2+2)(p_1-p_2)}{k_2^2 - 4}\right]$

$= \frac{1}{2}\left[\frac{k_2 p_1 + \cancel{k_2 p_2} - \cancel{2p_1} - 2p_2 + k_2 p_1 - \cancel{k_2 p_2} + \cancel{2p_1} - 2p_2}{k_2^2 - 4}\right]$

$$\boxed{\dot{q}_2 = \frac{k_2 p_1 - 2p_2}{k_2^2 - 4}}$$

$$\therefore \dot{q}_1 = \frac{1}{2}\left[\cancel{k_2 p_1} + k_2 p_2 - 2p_1 - \cancel{2p_2} - \cancel{k_2 p_1} + k_2 p_2 - 2p_1 + \cancel{2p_2}\right] \frac{1}{k_2^2 - 4}$$

$$\boxed{\dot{q}_1 = \frac{k_2 p_2 - 2p_1}{k_2^2 - 4}}$$

Replacing these in H:

$$H = \frac{\left[(k_2 p_2 - 2p_1)^2 + (k_2 p_1 - 2p_2)^2 + k(k_2 p_2 - 2p_1)(k_2 p_1 - 2p_2)\right]}{(k_2^2 - 4)^2} - k_1 q_1^2 \quad \text{(Ans)}$$

[$\because$ H is a f^n of q, p, t]

(4) p ces: (x, y).

$$\therefore p_x = \frac{\partial L}{\partial \dot{x}} = m\dot{x} + qA_x$$

$$\therefore p_y = \frac{\partial L}{\partial \dot{y}} = m\dot{y} + qA_y$$

$$\therefore H = \sum_i p_i \dot{q}_i - L$$

$$= (m\dot{x} + qA_x)\dot{x} + (m\dot{y} + qA_y)\dot{y} - L$$

$$= m\dot{x}^2 + qA_x\dot{x} + m\dot{y}^2 + qA_y\dot{y} - \frac{m}{2}(\dot{x}^2 + \dot{y}^2) - q(A_x\dot{x} + A_y\dot{y}) + q\phi(x, y)$$

$$\Rightarrow H = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) + q\phi(x, y)$$

$$= \frac{m}{2}\left[\left(\frac{p_x - qA_x}{m}\right)^2 + \left(\frac{p_y - qA_y}{m}\right)^2\right] + q\phi(x, y)$$

$$H(x, y, p_x, p_y) = \frac{1}{2m}\left[p_x^2 + p_y^2 + q^2A^2 - 2p_x qA_x - 2p_y qA_y\right] + q\phi(x, y)$$

(Ans)